

AI Executive Decision Engine for Last-Minute Productivity

Vibe2Ship Hackathon 2026 | Problem Statement 1: The Last-Minute Life Saver

GitHub Repository: <https://github.com/sharanyachilari01/DeadlineAI>

Deployed Application: <https://deadlineai-658ba.web.app/>

DeadlineAI

Document Purpose

This document explains DeadlineAI in detail for the Vibe2Ship hackathon submission. It covers the selected problem statement, solution overview, step-by-step usage, key features, architecture, AI logic, reasoning flow, technologies used, Google technologies utilized, testing, and future scope.

1. Problem Statement Selected

The selected Vibe2Ship problem statement is Problem Statement 1: The Last-Minute Life Saver.

The challenge is to build an AI-powered productivity companion that proactively assists users in planning, prioritizing, and completing tasks before deadlines are missed. The solution is expected to move beyond traditional reminders and focus on helping users take meaningful action.

The target users include students, professionals, entrepreneurs, and anyone managing multiple assignments, meetings, payments, interviews, personal commitments, and learning goals. The central problem is not merely that people forget tasks. The deeper problem is that people often do not know how to act when many commitments compete for limited time.

2. Understanding the Problem

Most reminder applications are passive. They tell the user that a deadline exists, but they do not help the user decide what action to take next. For example, a normal productivity tool may remind a student that an assignment is due tomorrow, but it does not answer whether the student should work on the assignment now, postpone a resume update, pay an urgent fee first, or split work into a realistic schedule.

The real-world difficulty appears when multiple deadlines overlap. A user may have a project submission due tonight, an interview tomorrow morning, a bill payment due today, and a personal commitment for tomorrow. In this situation, a list of reminders is not enough. The user needs decision support.

Core Problem

Existing tools remind users about deadlines. DeadlineAI helps users decide what to do, what to postpone, and how to recover when the schedule becomes overloaded.

3. Solution Overview

DeadlineAI is an AI-powered Executive Decision Engine. It acts like a personal chief of staff for the user's workload. Instead of only storing tasks, it analyzes the full task list together and produces an execution strategy for the day.

The user enters tasks with details such as title, description, deadline, estimated hours, importance, category, and progress. Separately, the user enters today's available focused hours. DeadlineAI combines these inputs and asks Google Gemini to reason about the workload holistically. The output includes a daily mission, prioritized tasks, a duration-based execution plan, proactive notices, rescue recommendations, what-if simulations, and productivity insights.

Traditional To-do App	DeadlineAI
Shows a list of tasks	Analyzes the complete workload
Sends reminders	Creates an actionable plan
User decides priority manually	AI recommends priority with reasoning
Often ignores available time	Uses daily focused hours to detect overload
No recovery plan	Provides Rescue Mode
No consequence prediction	Provides What-If simulation

4. How DeadlineAI Moves Beyond Traditional Reminders

From reminders to decisions

DeadlineAI does not stop at notifying the user. It recommends the best next action and explains why that action is better than alternatives.

From static lists to dynamic planning

The plan changes when tasks are added, completed, deleted, or when the user changes today's available focused hours.

From urgency-only sorting to contextual reasoning

The AI considers deadline, effort, progress, importance, category, and workload pressure together.

From panic to rescue strategy

When work exceeds available hours, Rescue Mode creates a minimum viable recovery plan and explicitly names tasks to postpone.

5. Step-by-Step User Guide

This section explains exactly how a user can use DeadlineAI from the first visit to task completion.

Step 1: Open the application

The user visits the deployed application link. The landing page introduces DeadlineAI and explains that it is not only a task manager but an AI Decision Engine that helps users execute missions.

Step 2: Sign up or log in

A new user creates an account using email and password. Existing users sign in. Authentication is handled using Firebase Authentication.

Step 3: Enter today's focused hours

After logging in, the user is asked how many focused hours are realistically available today. This value is stored as daily user context rather than inside every task. It helps the AI decide whether the workload is manageable, busy, overloaded, or critical.

Step 4: Add tasks

The user goes to Add Task and enters task details such as title, description, deadline, estimated hours, importance, progress, and category. These details are stored in Firestore under the authenticated user.

Step 5: View Mission Control

The dashboard shows the daily mission, first action, executive reasoning, proactive notices, prioritized workload, execution plan, and daily progress.

Step 6: Start the mission

The Start Mission button helps the user begin with the recommended first task. This turns the AI recommendation into an action-oriented workflow.

Step 7: Use Rescue Mode when overloaded

If the schedule is unrealistic, Rescue Mode recommends what to do immediately and which tasks should be postponed.

Step 8: Run What-If simulations

The user can select a task and scenario, such as skipping a task today, and DeadlineAI explains the likely consequences and a better alternative.

Step 9: Review Insights

The Insights page summarizes workload score, stress level, productivity score, focus window, and AI forecast.

Step 10: Complete or delete tasks

Users can mark tasks complete or delete tasks. The AI then recalculates the plan based on the updated workload.

6. Complete Feature Explanation

DeadlineAI was designed around one central idea: a productivity tool should not only store tasks; it should help the user decide what to do. Every major feature is connected to this goal. The application captures structured task information, combines it with the user's available productive hours, sends the workload to an executive AI layer, and then converts that reasoning into missions, plans, alerts, simulations, and insights.

User Input -> Firestore -> AI Context -> Decision Engine -> Gemini / Fallback -> Executive State -> Actionable UI

The following sections explain the implemented features in detail, including purpose, workflow, internal logic, AI reasoning, and practical benefit.

6.1 Secure User Authentication

Purpose

DeadlineAI begins with Firebase Authentication because the application deals with personal productivity information. A user may enter assignments, interview preparation, financial payments, family commitments, or personal errands. These items should not exist in a shared workspace. Authentication ensures every user has a separate account and a private planning environment.

User Workflow

- The user opens the deployed DeadlineAI web application.
- The user creates an account or signs in using email and password.
- After successful authentication, the user is redirected to Mission Control.
- Protected routes prevent unauthenticated users from accessing Dashboard, Add Task, Rescue Mode, What-If, and Insights.
- The navigation bar gives access to all core modules only after login.

Internal Logic

Firebase Authentication returns a unique user identifier after login. DeadlineAI uses this UID to scope Firestore data. Tasks and settings are therefore read and written only for the current account. This is important because the AI Decision Engine must reason about a single user's workload, not a mixed global dataset.

Layer	Responsibility
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Firestore Authentication	Verifies the user and manages the active session.
Auth Context	Stores the current user state in the React application.
Protected Route	Redirects unauthenticated users away from private pages.
Firestore UID Mapping	Ensures task and settings data are tied to the signed-in user.

Why this matters for AI

The AI recommendations become meaningful only when they are personalized. A student preparing for an interview and another user managing personal errands should receive completely different plans. Authentication makes this possible by giving the AI a consistent user-specific data boundary.

6.2 Daily Available Productive Hours

Purpose

Daily Available Productive Hours is one of the most important features in DeadlineAI. Most task managers ask what tasks exist, but they do not ask how much time the user realistically has today. This missing context causes unrealistic schedules. DeadlineAI solves this by asking the user for focused hours available for the current day.

Why it is global instead of task-level

The application originally could have treated available time as a field inside every task. That approach is unrealistic because availability belongs to the user's day, not to an individual task. The same assignment may exist for several days, but the user may have 6 hours today and only 2 hours tomorrow. Therefore, DeadlineAI stores available hours as daily user context.

Design Option	Problem	DeadlineAI Decision
Available time per task	Creates stale data and forces repeated editing.	Rejected.
Available time as daily user setting	Matches real-life availability and changes once per day.	Implemented.
No available time field	AI cannot know whether the workload is feasible.	Rejected.

AI Reasoning

The Decision Engine compares total remaining effort against available focused hours. This comparison affects mission generation, execution plan duration, stress level, rescue mode, notifications, and focus window recommendations. If

available hours are low, the system shifts from a full plan to a minimum viable plan. If available hours are high, it can recommend deeper work blocks and more complete execution.

Example

If a user has 10 hours of remaining tasks but only 4 focused hours today, DeadlineAI identifies overload. It will not pretend everything can be finished. It recommends immediate priorities and postponements.

User Benefit

The user receives realistic advice rather than generic advice. This makes DeadlineAI closer to an executive assistant than a reminder list because it evaluates capacity before recommending action.

6.3 Add Task Module

The Add Task module is the main information collection layer of DeadlineAI. Each field has a direct role in the AI reasoning process. The form intentionally collects more than a title and deadline because the system needs enough context to make useful decisions.

Field	Allowed / Expected Input	Purpose in AI Reasoning
Task Title	Short name of the task	Used in mission, cards, notifications, rescue plan, execution plan, and what-if outputs.
Description / Context	Free text explanation	Helps Gemini understand the task type, consequence, and real-world meaning.
Deadline	Date and time	Determines urgency, overdue status, risk level, and ordering.
Estimated Hours	Numeric effort estimate	Used to calculate workload, feasibility, execution blocks, stress, and rescue mode.
Importance	High, Medium, Low	Captures user preference, but AI can adjust based on urgency and effort.
Category	Work, Personal, Urgent Issue, Learning, and other supported categories used in the app	Improves contextual reasoning and wording.
Progress	0 to 100 percent	Helps AI distinguish unfinished tasks from nearly completed tasks.

Task Title

The title is the main label used across the system. It makes AI recommendations understandable because outputs can directly reference the task. For example, “Submit Semester Project” is clearer than “Task 1.” The title appears in priority cards, rescue recommendations, notifications, and what-if simulations.

Description / Context

The description helps the AI understand meaning. A title such as “Prepare Interview” is broad; the description may reveal that the user needs DSA revision, HR questions, or company research. Gemini uses this context to produce better tradeoffs and expected benefits.

Deadline

Deadline converts a task from a general intention into a time-bound commitment. The AI uses the deadline to detect overdue items, same-day urgency, tomorrow risk, and long-term flexibility. A small task due today can outrank a larger task due next week because missing it has an immediate consequence.

Estimated Hours

Estimated Hours is the effort signal. It allows DeadlineAI to calculate whether the workload fits inside available productive time. Without this field, Rescue Mode, stress analysis, execution planning, and success probability would be guesswork. The same deadline can produce different plans depending on whether the task takes 15 minutes or 5 hours.

Importance

Importance captures the user's own judgement: High, Medium, or Low. However, DeadlineAI does not blindly sort by importance. A medium-importance fee payment due today may become more urgent than a high-importance learning task due next week because the consequence is more immediate.

Category

Category improves the tone and reasoning of AI outputs. Academic tasks may affect grades, career tasks may affect opportunities, finance tasks may involve penalties, and personal tasks may affect relationships or well-being. This lets the AI write more human and specific recommendations.

Progress

Progress prevents the AI from treating all tasks equally. A task that is 80 percent complete can often be finished quickly and remove a large source of stress. A task at 0 percent may require more setup time even if it has the same deadline.

Internal Workflow

Form Input -> Validation -> Firestore Save -> AI Context Update -> Decision Engine Evaluation -> Dashboard Refresh

After the task is submitted, the raw task data is stored in Firestore. The AI does not permanently overwrite the task with reasoning. Instead, the AI Context merges task data with executive analysis while rendering the interface. This keeps Firestore as the source of truth while allowing AI recommendations to change dynamically.

6.4 Executive Dashboard / Mission Control

The Dashboard is the central workspace of DeadlineAI. It is intentionally called Mission Control because its purpose is not to show a passive list. It converts all available task data into a day-level decision strategy.

Dashboard Section	What it does	Reasoning behind it
Today's Mission	Shows the main objective for the day.	Prevents the user from starting randomly.
Executive Reasoning	Explains why the mission was selected.	Builds trust in the recommendation.
First Action	Gives the immediate next step.	Moves the user from planning into execution.
Executive Notices	Shows proactive warnings and opportunities.	Surfaces risks without making the user inspect every task.
Executive Prioritization	Displays all active tasks with AI enrichment.	Keeps tasks visible while giving order and explanation.
Today's Execution Plan	Allocates available time into duration blocks.	Turns priorities into a usable work plan.
Daily Progress	Shows completed versus total tasks.	Encourages completion and provides feedback.

Mission Logic

The mission is generated by considering the full workload. It may focus on clearing an irreversible deadline, reducing tomorrow's pressure, finishing an almost-complete project, or protecting time for a high-effort task. The mission is not always simply the first task in the list; it is the objective that best stabilizes the user's day.

Task Visibility Logic

The Dashboard shows all active Firestore tasks. This design was chosen after testing revealed that relying only on the AI's prioritizedTasks array could make non-prioritized tasks seem to disappear. The final implementation uses Firestore as the source of truth and AI output as enrichment. Every task remains visible unless completed or deleted.

Important Design Principle

AI may prioritize and explain tasks, but it should never hide user data. DeadlineAI preserves task visibility while still presenting an ordered executive plan.

Recalculate Priorities

The Recalculate Priorities action lets the user request a fresh evaluation. This is useful after adding multiple tasks, changing available hours, or testing different workloads. The Decision Engine uses caching to avoid unnecessary repeated calls when data has not changed.

6.5 Explainable Executive Prioritization

Executive Prioritization is the feature that turns DeadlineAI from a task manager into an AI decision system. The AI does not only assign a number. It explains urgency, risk, tradeoffs, and expected benefits.

Output on Task Card	Meaning
Rank	Where the task falls in the current execution order.
Risk Level	How dangerous it is to delay or ignore the task.
Score	A compact indicator of priority strength.
Next Action	The first practical action the user should take.
Tradeoffs and Consequences	What the user sacrifices or risks by focusing here or delaying it.
Expected Benefit	What improves if the user completes or advances the task.
Reasoning Panel	Additional explanation behind the recommendation.

AI Reasoning Factors

- Deadline proximity: today and overdue tasks receive urgent attention.
- Estimated effort: long tasks may require deep work; short tasks can become quick wins.
- Importance: high importance increases priority but does not override every other factor.
- Progress: nearly complete tasks may be recommended because they can remove an entire deadline quickly.
- Available hours: limited time changes the plan from completion to survival strategy.
- Category: finance, academic, career, personal, and learning tasks receive context-specific reasoning.

This explainability is important for user trust. A user is more likely to follow a recommendation when the system explains why it matters and what consequence is being avoided.

6.6 Proactive Notification Center

Notifications in DeadlineAI are designed as executive notices rather than normal reminders. They do not simply repeat that a deadline exists. They explain what the user should pay attention to and why it matters now.

Notice Type	Purpose	Example
Critical / Overload	Warns when the schedule is not feasible.	You have more work than available hours.
Quick Win	Highlights short tasks that can remove immediate pressure.	A 15-minute payment due today should be cleared first.
Safe Postponement	Identifies tasks that can wait.	A low-risk learning task due next week can be moved.
Mission Reminder	Keeps attention on the main objective.	The day's mission is still unfinished.
Opportunity	Suggests a beneficial action before it becomes urgent.	Start interview preparation early to reduce tomorrow's load.

Logic

The notification layer uses the executive state generated by the AI. It can also rely on fallback reasoning if the API is unavailable. The notices reference actual task names and user workload context rather than sounding generic.

6.7 Today's Execution Plan

The Execution Plan translates prioritization into action. The plan uses duration blocks such as 15 minutes, 60 minutes, or 90 minutes instead of fixed clock times. This was a deliberate trust-focused design decision.

Why duration-based planning is better

DeadlineAI does not currently know the user's calendar. If it displayed exact times such as 09:00 to 10:30, it would imply knowledge the system does not have. Duration-based planning tells the user how long to spend without pretending to know when they are free. This makes the plan flexible and realistic.

Block Type	Typical Duration	Purpose
Inbox and Planning	10-15 minutes	Set direction and reduce confusion.
Quick Win	15-45 minutes	Clear a short urgent task.
Deep Work	60-120 minutes	Make progress on effort-heavy tasks.
Break	10-20 minutes	Avoid unrealistic continuous work.
Buffer / Review	15-30 minutes	Handle small delays and final checks.

Generation Logic

The plan is constrained by available hours. If the user has only 1 hour, the plan becomes a minimum viable plan. If the user has 8 hours, it can contain deeper work and breaks. Low-priority tasks are not scheduled when high-risk tasks remain unfinished and time is limited.

6.8 Rescue Mode

Rescue Mode is one of the strongest features of DeadlineAI. It activates when the user's workload becomes unrealistic. Instead of telling the user to complete everything, it identifies what must be done, what can be postponed, and how to recover.

Rescue Component	Purpose
Immediate Action	The first move that reduces the highest risk.
Success Probability	A feasibility estimate if the rescue plan is followed.
Next 30 Minutes	A short actionable block to create momentum.
Next 2 Hours	A focused block for the highest-impact work.
Postpone Today	Explicit list of tasks to delay and why.
Reasoning	Explanation behind the survival strategy.

Postponement Logic

The AI chooses postponements by considering deadline distance, estimated hours, importance, progress, and category. Low-impact tasks due later are postponed before tasks that are due today, nearly complete, or high consequence. This makes Rescue Mode practical rather than motivational.

Example

If the user has a project due today, an interview tomorrow, and a learning video due next week, Rescue Mode should postpone the learning video and protect the project/interview tasks.

Why it matters

During overload, the most important decision is often what not to do today. Rescue Mode makes that decision explicit and reduces guilt by explaining why postponement is logical.

6.9 What-If Simulator

The What-If Simulator helps users test decisions before committing to them. This is useful when the user is considering skipping a task, delaying a task, or reducing effort.

Input / Output	Explanation
Selected Task	The task whose future is being tested.
Scenario	The user's possible decision, such as skipping or delaying.
Consequences	What may happen if the scenario is followed.
Risk Increase	How much the decision raises workload or deadline risk.
AI Confidence	How confident the system is in the prediction.
Better Alternative	A safer compromise recommended by the AI.

AI Logic

The simulator uses task title, deadline, estimated effort, progress, and the rest of the workload to predict consequences. For example, skipping a four-hour project due today is very different from skipping a low-priority learning task due next week.

User Benefit

This feature changes planning from reactive to predictive. Users can see the cost of a decision before it creates a real problem.

6.10 Executive Insights

Executive Insights provides a summary of workload health. It is not meant to be a decorative analytics page; it explains whether the day is realistic, what is causing stress, and where the user should focus first.

Insight	Logic
Workload Score	Reflects intensity of remaining work.
Stress Level	Classifies workload as Calm, Manageable, Busy, Overloaded, or Critical.
Productivity Score	Reflects completion progress and workload balance.
Focus Window	Suggests how long to focus first and on which task.
Forecast	Explains whether the current workload is feasible.
Reasoning Panel	Gives deeper explanation behind the summary.

Stress Level Reasoning

Stress level is based on the relationship between remaining work and available hours, along with urgent deadlines and overdue tasks. A light workload with no near deadline may be Calm. A full but realistic day may be Busy. A workload that exceeds available hours becomes Overloaded or Critical.

Focus Window Reasoning

The focus window is derived from the highest-value work, available time, and the execution plan. It avoids fake clock times and instead recommends durations such as the first 60 or 90 minutes. This keeps the recommendation realistic and flexible.

6.11 Offline AI Fallback and Quota Resilience

DeadlineAI is built to continue functioning when Gemini is temporarily unavailable. During testing, quota limits were encountered, so the fallback system became important for reliability. The application should not become useless just because an API call fails.

Failure Case	Fallback Response
Gemini quota exceeded	Use locally computed executive state.
Network issue	Generate local priorities and insights.
Malformed AI response	Use safe default structure.
Empty task list	Show calm empty state instead of crashing.
Old cached schema	Render safely with backward compatibility.

Why this is important

A hackathon demo or real user session can fail if a cloud AI API becomes unavailable. The fallback engine ensures that Dashboard, Rescue Mode, Insights, and What-If remain usable. This improves completeness and product reliability.

6.12 Real-Time Synchronization and Caching

DeadlineAI uses Firestore listeners and React Context to keep task state synchronized. When a user adds, completes, deletes, or updates a task, the interface receives the change and recalculates recommendations.

Caching Logic

To avoid unnecessary Gemini requests, the Decision Engine hashes the current task list and daily available hours. If the data has not changed, it can reuse cached executive state. This reduces API usage, improves speed, and helps avoid quota exhaustion.

Firestore Snapshot -> AIContext -> Hash Check -> Gemini or Cache/Fallback -> Executive State -> UI Refresh

User Benefit

The user does not need to manually rebuild the plan after each change. The application behaves like a live assistant that continuously adapts to the current workload.

6.13 Feature Interconnection Summary

Feature	Primary Input	Main Output
Authentication	Email and password	Private user workspace
Available Hours	Daily productive time	Capacity-aware planning
Add Task	Structured task fields	AI-ready workload data
Dashboard	Tasks + executive state	Mission and prioritized plan
Notifications	AI workload analysis	Proactive alerts
Execution Plan	Prioritized tasks + hours	Duration-based schedule
Rescue Mode	Overloaded workload	Recovery strategy
What-If	Selected task + scenario	Predicted consequence
Insights	All workload data	Executive summary
Fallback	Local task data	Resilient AI outputs
Caching	Task/settings hash	Reduced repeated API calls

7. AI Decision Engine: Logic and Reasoning

The **AI Decision Engine** is the intelligence layer of **DeadlineAI**. Unlike traditional task management applications that evaluate tasks independently, DeadlineAI analyzes the user's **entire workload as a unified decision-making problem**. Instead of prioritizing tasks using only deadlines or importance, the engine considers multiple factors simultaneously, including deadlines, estimated effort, available productive hours, task importance, completion progress, and contextual information.

Rather than answering the question **"Which task should be done first?"**, the AI answers:

"Given everything on the user's schedule today, what is the smartest and most realistic plan to maximize productivity while minimizing future risk?"

This holistic reasoning enables DeadlineAI to generate recommendations that are personalized, explainable, and practical for real-world productivity.

7.1 Inputs Used by the AI

To generate meaningful recommendations, the AI receives a complete representation of the user's workload rather than evaluating tasks one by one.

The information supplied includes:

- Active Tasks
- Task Titles
- Task Descriptions
- Deadlines
- Estimated Completion Time
- Importance Level
- Task Category
- Current Progress
- Daily Available Productive Hours
- Completion Status

Providing the complete workload enables the AI to understand relationships between tasks. It can identify conflicts, detect workload overload, estimate future pressure, and generate recommendations that optimize the entire day rather than individual tasks.

7.2 Prioritization Logic

DeadlineAI **does not simply sort tasks according to deadlines.**

Instead, the AI evaluates multiple decision factors simultaneously to determine the most effective execution order.

These factors include:

- Deadline proximity
- Estimated effort
- Current completion percentage

- User-defined importance
- Consequences of delay
- Available productive hours
- Overall workload distribution

For example, a **30-minute scholarship application due today** may receive higher priority than a **five-hour assignment due tomorrow** because completing the shorter task immediately removes irreversible risk while requiring minimal effort.

This multi-factor reasoning produces recommendations that closely resemble real executive decision-making instead of simple chronological sorting.

7.3 Risk Level Logic

Every prioritized task receives a **dynamic Risk Level** that indicates how urgently it should be completed.

Possible classifications are:

Critical – Immediate action is required to avoid severe consequences.

High Risk – Delaying the task is likely to create significant problems in the near future.

Medium Risk – Delay is acceptable for a short period but should not continue indefinitely.

Low Risk – The task can safely be postponed without major consequences.

Risk levels are calculated using:

- Deadline proximity
- Remaining effort
- Completion progress
- Available productive hours
- Overall workload pressure

This allows users to immediately recognize which tasks demand attention and which tasks have scheduling flexibility.

7.4 Tradeoffs and Consequences

Unlike conventional productivity applications, DeadlineAI explains the **trade-offs** associated with every important recommendation.

Instead of simply saying "*Complete Task A first,*" the AI explains:

- What benefits the user gains by working on that task immediately.
- Which other activities may temporarily receive less attention.
- What negative consequences could occur if the recommendation is ignored.

These explanations make the decision-making process transparent and significantly improve user trust in the AI recommendations.

7.5 Expected Benefits

For every prioritized task, DeadlineAI predicts the positive outcome of completing the task immediately.

Examples include:

- Reducing tomorrow's workload
- Avoiding missed deadlines
- Improving interview preparedness

Displaying expected benefits alongside trade-offs enables users to understand both the risks and rewards of every recommendation.

7.6 Stress Level Logic

Instead of estimating stress using a single numerical value, DeadlineAI evaluates the relationship between **workload demand** and **available productive capacity**.

Stress is classified into five categories:

- Calm
- Manageable
- Busy

- Overloaded
- Critical

The classification considers:

- Total remaining workload
- Daily available productive hours
- Number of urgent tasks
- Number of overdue tasks
- Remaining work after considering task progress

Rather than displaying only the category, the AI generates a natural language explanation describing exactly why the workload has been classified at that level.

7.7 Focus Window Logic

DeadlineAI intentionally avoids assigning fixed calendar times because users have different daily schedules.

Instead, the AI recommends flexible **duration-based focus windows**, such as:

- First 30 Minutes
- First 60–90 Minutes
- Next Deep Work Block
- Final Review Session

The recommended duration depends on:

- Priority of the selected task
- Estimated effort
- Remaining workload
- Available productive hours
- Overall schedule pressure

This approach makes the recommendations adaptable to each user's daily routine while still encouraging structured work sessions.

7.8 What-If Reasoning

The **What-If Simulator** allows users to explore the consequences of changing their schedule before making a decision.

When a task is skipped, delayed, or postponed, the AI predicts:

- Expected Risk Increase
- Likely Consequences
- AI Confidence
- Better Alternative Strategy

Instead of simply warning users, the simulator provides practical guidance that helps them make informed scheduling decisions.

7.9 Fallback Reasoning

To ensure uninterrupted functionality, DeadlineAI includes a deterministic **Offline Fallback Engine**.

If the Gemini API becomes unavailable because of quota limitations, connectivity issues, or temporary service interruptions, the application automatically generates executive recommendations locally.

The fallback engine produces:

- Task Prioritization
- Rescue Recommendations
- Execution Plans
- Executive Notifications
- Productivity Insights

Although the fallback engine does not provide the same depth of reasoning as Gemini, it guarantees that the application remains fully functional under all conditions.

8. System Architecture and Data Flow



The architecture separates raw task data from AI-generated reasoning. Firestore remains the source of truth for tasks. The AI enriches those tasks with risk level, priority score, reasoning, and recommendations. This prevents tasks from disappearing if the AI returns only a subset of prioritized items.

9. Firestore Data Model

9.1 User Tasks

Path: users/{userId}/tasks/{taskId}

- title
- description
- deadline
- estimatedHours
- importance
- category
- progress
- completed
- createdAt / updatedAt

9.2 Daily User Settings

Path: users/{userId}/settings/daily

- availableHoursToday
- lastUpdatedDate

Daily availability is stored separately from tasks because available time changes every day. This is an important data-modeling decision that makes replanning realistic.

10. Technologies Used

Technology	Purpose
React.js	Used to build the component-based frontend interface.
Vite	Used for fast development and production builds.
Tailwind CSS	Used for responsive dark-themed UI styling.
React Router	Used for navigation and protected routes.
Lucide React	Used for clean consistent icons.
JavaScript	Used for app logic, Firebase integration, and AI service handling.
Git and GitHub	Used for version control and public source code hosting.

11. Google Technologies Utilized

Google Gemini 2.5 Flash

Used for executive reasoning, workload analysis, mission generation, prioritization, rescue planning, what-if simulation, insights, and explainable AI outputs.

Firebase Authentication

Used for secure email/password sign up, login, logout, and protected user sessions.

Cloud Firestore

Used as the cloud database for user-scoped tasks and daily settings.

Firebase Hosting

Used to deploy the React application publicly on Google Cloud infrastructure.

Google AI Studio

Used to obtain and configure the Gemini API key for the application.

12. Testing and Edge Cases

DeadlineAI was tested with normal workloads, overloaded workloads, light workloads, same-deadline tasks, single-task cases, empty dashboard cases, and mixed personal-academic workloads. The purpose was to verify that the AI remains useful across realistic scenarios and not only for one predefined demo.

Test Case	Purpose
Normal Student Workload	Checks whether DBMS, groceries, aptitude preparation, family call, and learning tasks are prioritized logically.
Massive Overload	Checks whether the app detects insufficient available hours and recommends postponement.
Almost No Urgency	Checks whether the app avoids creating fake panic when the workload is manageable.
Same Deadline Tasks	Checks whether the app compares effort and importance instead of blindly sorting by deadline.
Single Task	Checks whether the app avoids unnecessary complexity when only one task exists.
Empty Dashboard	Checks that the app does not crash and shows a clear empty state.
Mixed Real-Life Workload	Checks academic, finance, career, personal, and learning categories together.

13. Alignment with Evaluation Matrix

Problem Solving & Impact

DeadlineAI directly addresses missed deadlines and decision fatigue by giving users an actionable plan.

Agentic Depth

The app generates missions, rescue plans, postponement recommendations, what-if simulations, and explanations.

Innovation & Creativity

The executive assistant framing goes beyond reminders and simple priority sorting.

Usage of Google Technologies

Uses Gemini, Firebase Authentication, Cloud Firestore, and Firebase Hosting.

Product Experience & Design

The product has a dark polished interface, task flows, dashboards, and responsive screens.

Technical Implementation

Uses React, Firebase, Firestore, protected routes, context providers, caching, and fallback logic.

Completeness & Usability

The app is deployed, usable, and supports end-to-end task management and AI planning.

14. Challenges Faced and Improvements Made

- Avoiding generic AI responses and making reasoning task-specific.
- Replacing fake clock-based timelines with duration-based execution plans.
- Moving available hours from task-level data to user daily context.
- Ensuring all tasks remain visible even if Gemini prioritizes only a subset.
- Handling Gemini quota limits through offline fallback.
- Fixing Firestore document path issues for user settings.
- Improving stress level categories and focus window reasoning.

15. Future Scope

- Google Calendar integration to convert duration blocks into real calendar slots.
- Gmail integration to detect deadlines from emails.
- Voice-enabled task entry and mission briefings.
- Mobile app version with push notifications.
- Habit and goal tracking for long-term productivity.
- Team collaboration for shared deadlines and group projects.
- Weekly productivity reports and trend analysis.

16. Conclusion

DeadlineAI transforms productivity from passive reminders into proactive decision-making. It helps users understand their workload, prioritize intelligently, recover from overload, and act before deadlines are missed. By combining Google Gemini with Firebase Authentication, Cloud Firestore, and Firebase Hosting, DeadlineAI demonstrates how AI can improve productivity by helping users make better decisions and complete tasks more effectively.